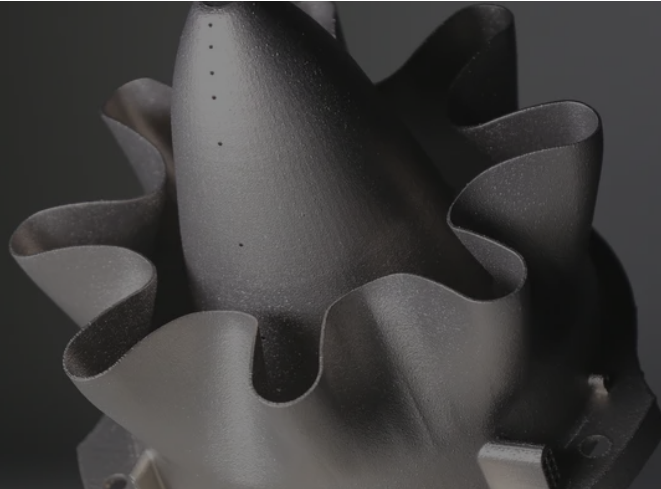


METAL SOLUTIONS

EOS FeNi36

Material Data Sheet



EOS FENI36

Nickel-Iron alloy with low coefficient of thermal expansion

EOS FeNi36 is a nickel-iron alloy characterized by low coefficient of thermal expansion within the temperature range of 30 to 150 °C.

MAIN CHARACTERISTICS

- Very low coefficient of thermal expansion (30-150 °C)
- Excellent dimensional stability under thermal cycling

TYPICAL APPLICATIONS

- Housings and structures for optical and scientific instruments
- Aerospace and satellite structures
- Other applications requiring low thermal expansion

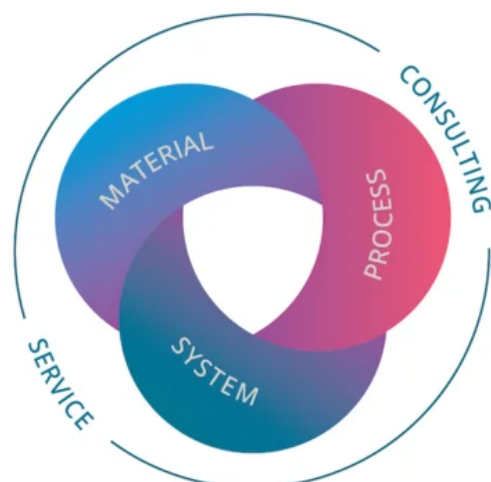
The EOS Quality Triangle

EOS uses an approach that is unique in the AM industry, taking each of the three central technical elements of the production process into account: the system, the material and the process. The data resulting from each combination is assigned a Technology Readiness Level (TRL) which makes the expected performance and production capability of the solution transparent.

EOS incorporates these TRLs into the following two categories:

- Premium products (TRL 7-9): offer highly validated data, proven capability and reproducible part properties.
- Core products (TRL 3 and 5): enable early customer access to newest technology still under development and are therefore less mature with less data.

All of the data stated in this material data sheet is produced according to EOS Quality Management System and international standards



POWDER PROPERTIES

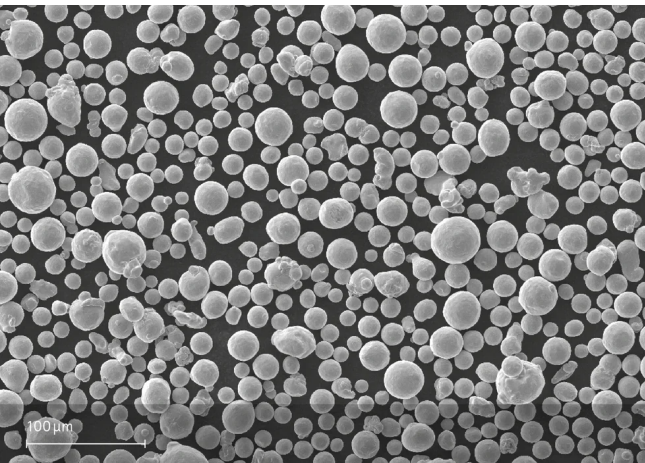
Material according to ASTM-F1684-06.

Powder Chemical Composition (wt.-%)

Element	Min.	Max.
Fe		Balance
Ni	35.5	37
Si	-	0.4
Mn	-	0.6
Cr	-	0.25
Co	-	0.1
C	-	0.05
Al	-	0.1
Mg	-	0.1
Zr	-	0.1
Ti	-	0.1
P	-	0.015
S	-	0.015

Powder Particle Size

Generic Particle Size Distribution	15 - 53 µm
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SEM image of EOS FeNi36 powder

HEAT TREATMENT

Description

Anneal the part in a non-oxidizing atmosphere for a minimum of 1 h at 875 °C +/-25 °C when thoroughly heated.

Cool at a rate not exceeding 300°C/h.

EOS FeNi36 for EOS M 290 | 40 µm & 80 µm

EOS M 290 - 40 µm - TRL 3

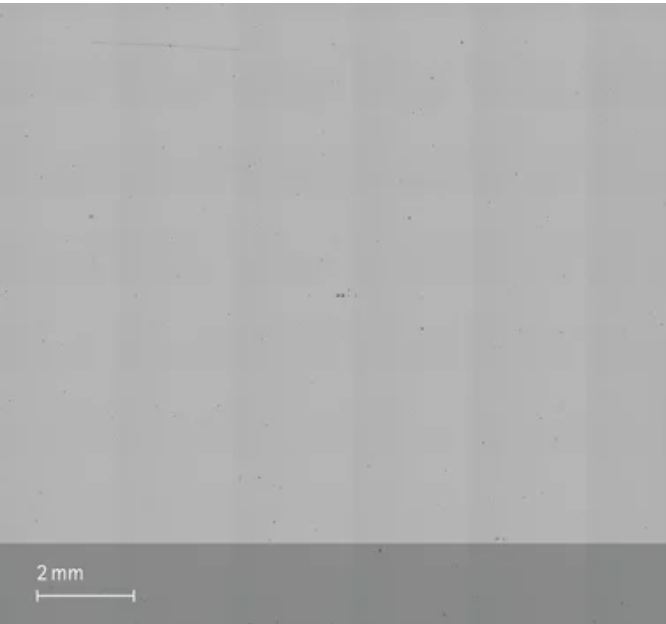
Process Information Metal

This process product is optimized for flexible and fast production of EOS FeNi36 parts with the EOS M 290 system. The parameter set has two different layer thickness options that can all be utilized within the same build: 40 µm and 80 µm. The 40 µm parameter set is ideal for parts needing great detail resolution and more dense structure. The 80 µm parameter set offers a build rate that is more than double that of the 40 µm parameter set. The parameter sets are assigned to different sections in the same build job depending on the requirements.

System Setup	EOS M 290
EOS Material set	FeNi36_040_080_CoreM291
Required Software Plan	EOS BUILD
Software Requirements	EOSPRINT 2.13 or newer EOSYSTEM 2.17 or newer
Recoater Blade	Ceramic
Nozzle	EOS Grid Nozzle
Inert gas	Argon
Sieve	75 µm

Additional Information	
Layer Thickness	40 µm, 80 µm
Volume Rate	3.7-7.9 mm³/s
Typical Dimensional Change after HT [%]	0.0 %

Chemical and Physical Properties of Parts



Etched micrograph, as manufactured

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	40 μm	0.05 %	-

Mechanical Properties

Mechanical Properties As Manufactured

EN ISO 6892-1 Room Temperature 40 μm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	340	420	40	78	-	-
Horizontal	380	460	35	80	-	-

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 40 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	310	420	42	78	-	-
Horizontal	320	460	36	78	-	-

Anneal the part in a non-oxidizing atmosphere for a minimum of 1 h at 875 °C +/-25 °C when thoroughly heated. Cool at a rate not exceeding 300°C/h.

Mechanical Properties Heat Treated

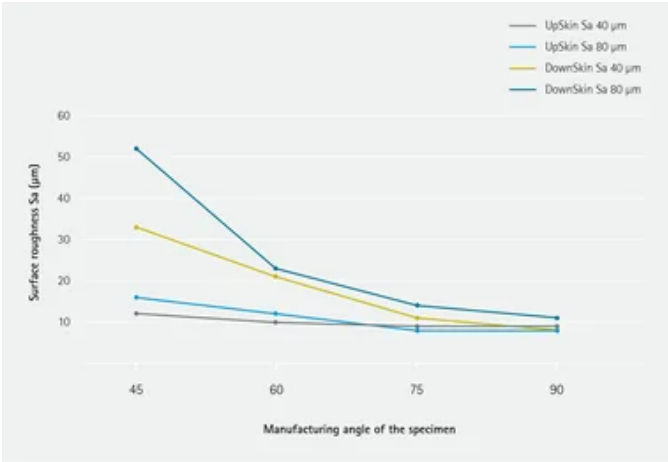
EN ISO 6892-1 Room Temperature 80 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	295	430	40	78	-	-
Horizontal	305	450	37	80	-	-

Anneal the part in a non-oxidizing atmosphere for a minimum of 1 h at 875 °C +/-25 °C when thoroughly heated. Cool at a rate not exceeding 300°C/h.

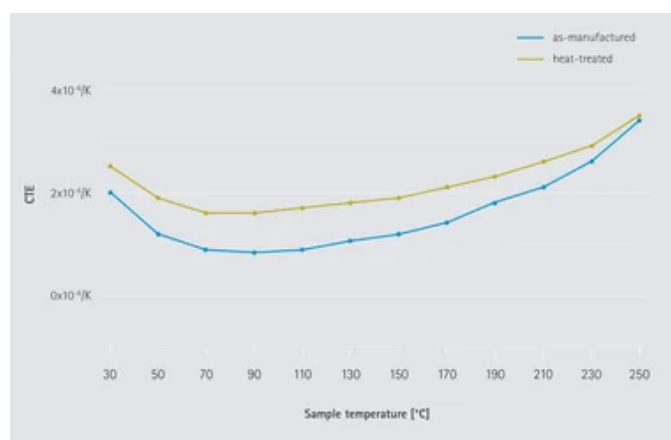
Hardness

Value	140
Unit	HV

Surface Roughness



Coefficient of Thermal Expansion



HEADQUARTERS

EOS GmbH Electro Optical Systems

Robert-Stirling-Ring 1
82152 Krailling / Munich
Germany

Tel.: +49 89 893 36-0
Email: info@eos.info
URL: www.eos.info

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Status as of 18.08.2026. Subject to technical modifications. EOS is certified according to ISO 9001.

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